

LOGIC AND PROOFS

Rules of Inference

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OUTLINE

1. RULES OF INFERENCE FOR PROPOSITIONAL LOGIC
2. RULES OF INFERENCE
3. USING RULES OF INFERENCE TO BUILD ARGUMENTS
4. RULES OF INFERENCE FOR QUANTIFIED STATEMENTS

Rules of Inference for Propositional Logic

Definitions:

- **Argument:** A sequence of propositions $p_1, p_2, \dots, p_n$ (also known as **premises**) such that $p_1 \wedge p_2 \wedge \dots \wedge p_n \rightarrow q$.
- All but the final proposition in the argument are called premises and the final proposition is called the **conclusion**.
- **Valid argument:** mean that the conclusion, or final statement of the argument, must follow from the truth of the preceding statements, or premise.
- **Rules of inference** are our basic tools for establishing the truth of statements.

Valid Arguments in Propositional Logic

Consider the following argument involving propositions (which, by definition, is a sequence of propositions):

“If you have a current password, then you can log onto the network.”

“You have a current password.”

Therefore,

“You can log onto the network.”

- Let p represent “You have a current password.”, and q “You can log onto the network.”. Then the argument has the form

$$\begin{array}{c} p \rightarrow q \\ p \\ \hline \therefore q \end{array}$$

Other examples of modus ponens

- If the cake is made with sugar, then the cake is sweet. The cake is made with sugar. Therefore, the cake is sweet.
- If Sam was born in Canada, then he is Canadian. Sam was born in Canada. Therefore, Sam is Canadian.
- If today is Tuesday, then John will go to work. Today is Tuesday. Therefore, John will go to work.

An **Inference Rule** is a pattern establishing that if we know that a set of premise statements of certain forms are all true, then we can validly deduce that a certain related conclusion statement is true.

Rules of Inference.					
<i>Rule of Inference</i>	<i>Tautology</i>	<i>Name</i>	<i>Rule of Inference</i>	<i>Tautology</i>	<i>Name</i>
$\begin{array}{l} p \\ p \rightarrow q \\ \hline \therefore q \end{array}$	$(p \wedge (p \rightarrow q)) \rightarrow q$	Modus ponens	$\begin{array}{l} p \\ \hline \therefore p \vee q \end{array}$	$p \rightarrow (p \vee q)$	Addition
$\begin{array}{l} \neg q \\ p \rightarrow q \\ \hline \therefore \neg p \end{array}$	$(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$	Modus tollens	$\begin{array}{l} p \wedge q \\ \hline \therefore p \end{array}$	$(p \wedge q) \rightarrow p$	Simplification
$\begin{array}{l} p \rightarrow q \\ q \rightarrow r \\ \hline \therefore p \rightarrow r \end{array}$	$((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$	Hypothetical syllogism	$\begin{array}{l} p \\ q \\ \hline \therefore p \wedge q \end{array}$	$((p) \wedge (q)) \rightarrow (p \wedge q)$	Conjunction
$\begin{array}{l} p \vee q \\ \neg p \\ \hline \therefore q \end{array}$	$((p \vee q) \wedge \neg p) \rightarrow q$	Disjunctive syllogism	$\begin{array}{l} p \vee q \\ \neg p \vee r \\ \hline \therefore q \vee r \end{array}$	$((p \vee q) \wedge (\neg p \vee r)) \rightarrow (q \vee r)$	Resolution

Example

State which rule of inference is used in the argument:

It is either hotter than 40^0C today or the pollution is dangerous. It is less than 40^0C degrees outside today. Therefore, the pollution is dangerous.

Solution

Let:

p : "It is hotter than 40^0C today,"

q : "The pollution is dangerous,"

Then this argument is of the form:

$$\begin{array}{c} p \vee q \\ \neg p \\ \hline \therefore q \end{array}$$

i.e. $(p \vee q) \wedge \neg p \rightarrow q$

This is an argument that uses the **Disjunctive Syllogism rule**.

Example

State which rule of inference is used in the argument:

If it rains today, then we will not have a barbecue today. If we do not have a barbecue today, then we will have a barbecue tomorrow. Therefore, if it rains today, then we will have a barbecue tomorrow.

Solution

Let:

p : "It is raining today,"

q : "We will not have a barbecue today,"

r : "We will have a barbecue tomorrow."

Then this argument is of the form:

$$\begin{array}{l} p \rightarrow q \\ q \rightarrow r \\ \hline \therefore p \rightarrow r \end{array}$$

Hence, this argument is a **hypothetical syllogism**.

Fallacies

Fallacies resemble rules of inference, but are based on contingencies rather than tautologies.

Using Rules of Inference to Build Arguments

Example

Show that the premises “It is not sunny this afternoon and it is colder than yesterday,” “We will go swimming only if it is sunny,” “If we do not go swimming, then we will take a canoe trip,” and “If we take a canoe trip, then we will be home by sunset” lead to the conclusion “We will be home by sunset.”

Solution

Let

p : It is sunny this afternoon.

q : It is colder than yesterday.

r : We will go swimming.

s : We will take a canoe trip.

t : We will be home by sunset.

Premises become

$\neg p \wedge q, r \rightarrow p, \neg r \rightarrow s, s \rightarrow t$. The conclusion is t .

Solution - Continuation

Step	Reason
1. $\neg p \wedge q$	Premise
2. $\neg p$	Simplification using (1)
3. $r \rightarrow p$	Premise
4. $\neg r$	Modus tollens using (2) and (3)
5. $\neg r \rightarrow s$	Premise
6. s	Modus ponens using (4) and (5)
7. $s \rightarrow t$	Premise
8. t	Modus ponens using (6) and (7)

Rules of Inference for Quantified Statements

The basic theme of these is that we can jump between a quantified statement (like $\exists k(n = 2k)$) and a specific instance or back.

Rules of Inference for Quantified Statements.	
Rule of Inference	Name
$\frac{\forall x P(x)}{\therefore P(c)}$	Universal instantiation
$\frac{P(c) \text{ for an arbitrary } c}{\therefore \forall x P(x)}$	Universal generalization
$\frac{\exists x P(x)}{\therefore P(c) \text{ for some element } c}$	Existential instantiation
$\frac{P(c) \text{ for some element } c}{\therefore \exists x P(x)}$	Existential generalization

Universal instantiation says that we can assume that an arbitrary $P(c)$ exists if we know something holds for all $P(x)$.

Universal generalization says that if we know $P(c)$ is true for an arbitrary c , it's true for all c .

Existential instantiation tells us if $P(x)$ is true for some value of x , there is a $P(c)$ for which it is true.

And **Existential generalization** tells us that if $P(c)$ is true for some c , it's true for at least one case.

Example

What rules of inference are used in this famous argument? “All men are mortal. Socrates is a man. Therefore, Socrates is mortal.”

$M(x)$ = “x is mortal”

$N(x)$ = “x is a man”

Premise 1	“All men are mortal”	$\forall x(N(x) \rightarrow M(x))$
Premise 2	“Socrates is a man”	$N(Socrates)$
Conclusion	“Socrates is mortal”	$M(Socrates)$

Step

1. $\forall x(N(x) \rightarrow M(x))$
 2. $N(Socrates) \rightarrow M(Socrates)$
 3. $N(Socrates)$
- $\therefore M(Socrates)$

Reason

- Premise 1
- Universal Instantiation from (1)
- Premise 2
- Modus Ponens from (2) and (3)

Example

What rules of inference are used in this argument? “No man is an island. Manhattan is an island. Therefore, Manhattan is not a man.”

$P(x)$ = “x is a man”

$Q(x)$ = “x is an island”

Premise 1 “No man is an island”

$\forall x(P(x) \rightarrow \neg Q(x))$

Premise 2 “Manhattan is an island”

$Q(Manhattan)$

Conclusion “Manhattan is not a man”

$\neg P(Manhattan)$

Step

1. $\forall x(P(x) \rightarrow \neg Q(x))$

2. $P(Manhattan) \rightarrow \neg Q(Manhattan)$

3. $Q(Manhattan)$

$\therefore \neg P(Manhattan)$

Reason

Premise 1

Universal Instantiation from (1)

Premise 2

Modus Tollens from (2) and (3)

OR

Premise 1	"No man is an island"	$\neg \exists x(M(x) \wedge I(x))$
Premise 2	"Manhattan is an island"	$I(Manhattan)$
Conclusion	"Manhattan is not a man"	$\neg M(Manhattan)$

Step

1. $\neg \exists x(M(x) \wedge I(x))$
 2. $\neg M(Manhattan) \vee \neg I(Manhattan)$
 3. $I(Manhattan)$
- $\therefore \neg M(Manhattan)$

Reason

- Premise 1
- Universal Instantiation from (1)
- Premise 2
- Disjunctive Syllogism from (2) and

(3)

Here's the form of the argument. $\neg \exists x(M(x) \wedge I(x))$.

$I(Manhattan)$. Therefore $\neg M(Manhattan)$.

The first statement is equivalent to $\forall x(\neg M(x) \vee \neg I(x))$.

Using universal instantiation, we get $\neg M(Manhattan) \vee \neg I(Manhattan)$.

Then, using disjunctive syllogism on that and the statement $I(Manhattan)$, we conclude $\neg M(Manhattan)$.

Example

Show that the premises “Everyone in this discrete mathematics class has taken a course in computer science” and “Marla is a student in this class” imply the conclusion “Marla has taken a course in computer science.”

Solution

Let

$D(x)$: x is in this discrete mathematics class.

$C(x)$: x has taken a course in computer science.

Then the premises are $\forall x(D(x) \rightarrow C(x))$ and $D(Marla)$. The conclusion is $C(Marla)$.

The following steps can be used to establish the conclusion from the premises.

Step

1. $\forall x(D(x) \rightarrow C(x))$
2. $D(Marla) \rightarrow C(Marla)$
3. $D(Marla)$
4. $C(Marla)$

Reason

Premise
Universal instantiation from (1)
Premise
Modus ponens from (2) and (3)

Example

Show that the premises to construct a valid argument showing that the conclusion “Someone who passed CSI131 did not read the notes” follows from “A student in CSI131 has not read the notes” and “Everyone in CSI131 passed the course.”

Solution Let

$D(x)$: x is a student in CSI131.

$B(x)$: x read the notes.

$P(x)$: x passed CSI131.

Using predicate logic, and starting on “A student in CSI131 has not read the notes” $\exists x(D(x) \wedge \neg B(x))$ and $\forall x(D(x) \rightarrow P(x))$. The conclusion is $\exists x(P(x) \wedge \neg B(x))$.

The following steps can be used to establish the conclusion from the premises.

Example

Show that the premises to construct a valid argument showing that the conclusion “Someone who passed CSI131 did not read the notes” follows from “A student in CSI131 has not read the notes” and “Everyone in CSI131 passed the course.”

Solution Let

$D(x)$: x is a student in CSI131.

$B(x)$: x read the notes.

$P(x)$: x passed CSI131.

Using predicate logic, translating, we have:

$\exists x(D(x) \wedge \neg B(x))$ Premise 1

$\forall x(D(x) \rightarrow P(x))$ Premise 2

$\therefore \exists x(P(x) \wedge \neg B(x))$ Conclusion

The following steps can be used to establish the conclusion from the premises.

Example - continuation

Step	Reason
(1) $\exists x(D(x) \wedge \neg B(x))$	Premise
(2) $D(a) \wedge \neg B(a)$	Existential instantiation from (1).
(3) $D(a)$	Simplification from (2)
(4) $\forall x(D(x) \rightarrow P(x))$	Premise
(5) $D(a) \rightarrow P(a)$	Universal instantiation from (4)
(6) $P(a)$	Modus polens from (3) and (4)
(7) $\neg B(a)$	Simplification from (2)
(8) $P(a) \wedge \neg B(a)$	Conjunction from (6) and (7)
(9) $\exists x(P(x) \wedge \neg B(x))$	Existential generalization from (8)

Example:

For these arguments, explain which rules of inference are used for each step.

“Linda, a student in this class, owns a red convertible. Everyone who owns a red convertible has gotten at least one speeding ticket. Therefore, someone in this class has gotten a speeding ticket.”

$c(x)$: x is in this class.

$r(x)$: x owns a red convertible.

$t(x)$: x has gotten a speeding ticket.

- | | |
|---------------------------------------|--------------------------------------|
| 1. $\forall x(r(x) \rightarrow t(x))$ | Hypothesis |
| 2. $r(Linda) \rightarrow t(Linda)$ | Universal instantiation using (1) |
| 3. $r(Linda)$ | Hypothesis |
| 4. $t(Linda)$ | Modus ponens using (2) and (3) |
| 5. $c(Linda)$ | Hypothesis |
| 6. $r(Linda) \wedge c(Linda)$ | Conjunction using (4) and (5) |
| 7. $\exists x(c(x) \wedge t(x))$ | Existential generalization using (6) |

Universal Modus Ponens

$$\forall x(P(x) \rightarrow Q(x))$$

$P(a)$, where a is a particular element in the domain

$$\therefore Q(a)$$

Example:

Assume that “For all positive integers n , if n is greater than 4, then n^2 is less than 2^n ” is true. Use universal modus ponens to show that $100^2 < 2^{100}$.

Solution

Let $P(n)$ denote “ $n > 4$ ” and $Q(n)$ denote “ $n^2 < 2^n$ ”. The statement “For all positive integers n , if n is greater than 4, then n^2 is less than 2^n ” can be represented by $\forall n(P(n) \rightarrow Q(n))$ where the domain consists of all positive integers. Assuming $\forall n(P(n) \rightarrow Q(n))$ is true. Note that $P(100)$ is true because $100 > 4$. It follows by universal modus ponens that $Q(100)$ is true, namely, that $100^2 < 2^{100}$.

Universal Modus Tollens

$$\begin{array}{l} \forall x(P(x) \rightarrow Q(x)) \\ \neg Q(a), \text{ where } a \text{ is a particular element in the domain} \\ \hline \therefore \neg P(a) \end{array}$$

The rule of universal instantiation can be combined with modus ponens to obtain the valid form of argument called universal modus ponens.

Example:

Rewrite the following argument using quantifiers, variables, and predicate symbols. Write the major premise in conditional form. Is this argument valid? Why?

All human beings are mortal

- Zeus is not mortal.
- Zeus is not human.

Solution

The major premise can be rewritten as

$\forall x$, if x is human then x is mortal.

Let

$H(x)$: x is human.

$M(x)$: x is mortal.

Z : Zeus.

The argument becomes.

$\forall x$, if $H(x)$ then $M(x)$.

$\neg M(x)$

$\neg H(x)$

This argument has the form of universal modus tollens and is therefore valid.

The End